EXPERIMENT 5

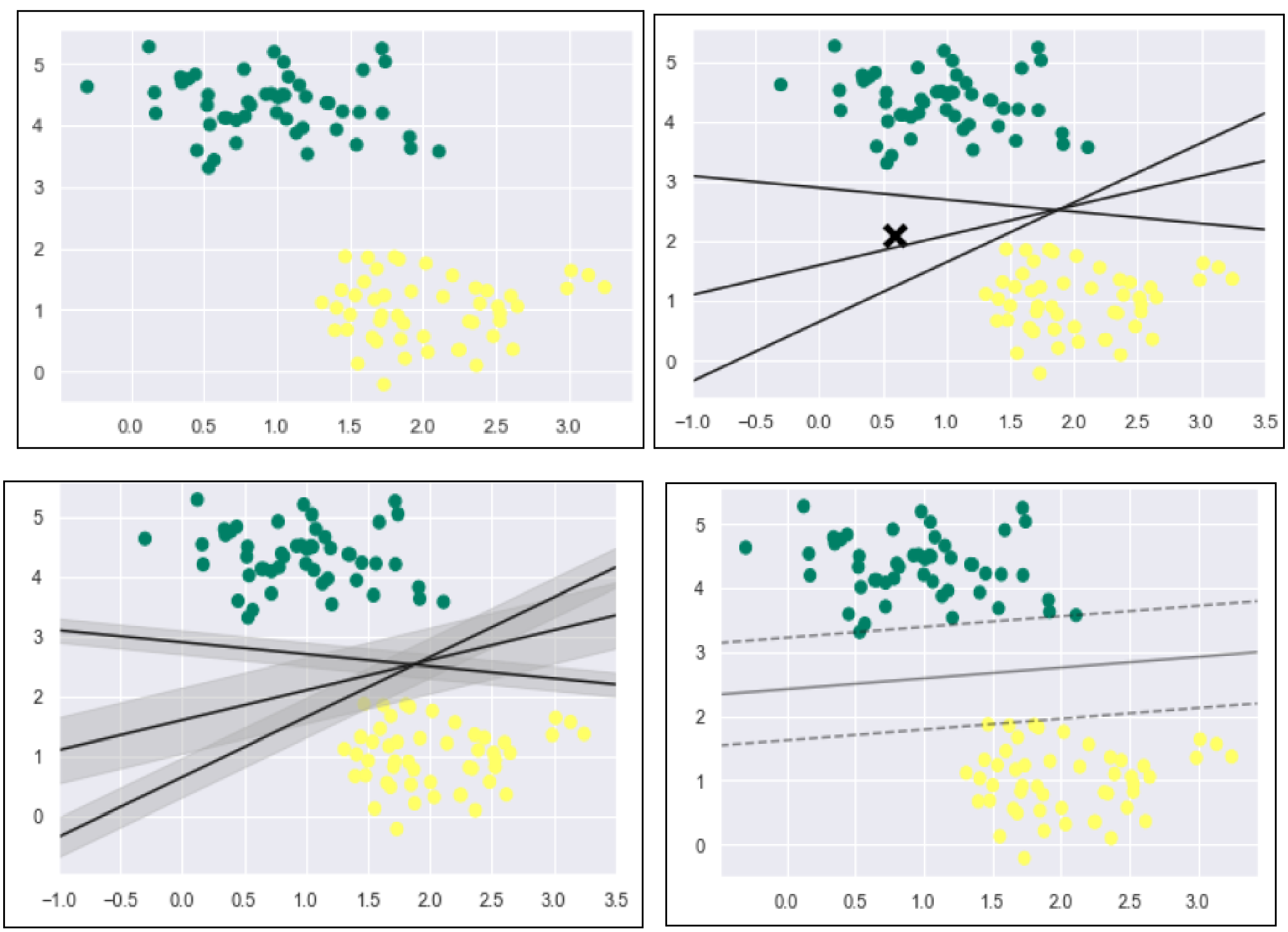
Implement Support Vector Machine (SVM) for classification with hyperparameter tuning

Support Vector Machine (SVM)

Support vector machines (SVMs) are powerful yet flexible supervised machine learning algorithms which are used for both classification and regression. But generally, they are used in classification problems. In the 1960s, SVMs were first introduced but later they got refined in 1990 also. SVMs have their unique way of implementation as compared to other machine learning algorithms. Nowadays, they are extremely popular because of their ability to handle multiple continuous and categorical variables.

It tries to find the best boundary known as a hyperplane that separates different classes in the data. It is useful when you want to do binary classification like spam vs. not spam or cat vs. dog.

Working of SVM :



1.Dataset Source

Dataset Name: Pima Indians Diabetes Dataset
Source: Kaggle
Link:
https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

File Used: diabetes.csv

This dataset is widely used for binary classification problems.

2.Dataset Description

Objective :
To predict whether a patient has diabetes based on medical measurements.
This is a binary classification problem.

Dataset Details

- Total Records: 768 samples
- Features: 8 numerical features
- Target Variable: Outcome

Features :

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Skin fold thickness
Insulin	Serum insulin level
BMI	Body Mass Index
DiabetesPedigreeFunction	Genetic influence factor
Age	Age of patient

Target Variable

Outcome:

- 0 → No Diabetes
- 1 → Diabetes

3.Mathematical Computation of SVM

Consider a binary classification problem with two classes, labeled as +1 and -1. We have a training dataset consisting of input feature vectors X and their corresponding class labels Y. The equation for the linear hyperplane can be written as:

$$w^T x + b = 0$$

Where:

- w is the normal vector to the hyperplane (the direction perpendicular to it).
- b is the offset or bias term representing the distance of the hyperplane from the origin along the normal vector w.

Distance from a Data Point to the Hyperplane

The distance between a data point $x_i$and the decision boundary can be calculated as:

$$d_i = \frac{w^T x_i + b}{||w||}$$

where ||w|| represents the Euclidean norm of the weight vector w.

Linear SVM Classifier

Distance from a Data Point to the Hyperplane:

$$\hat{y} = \begin{cases} 1 & : \ w^T x + b \geq 0 \\ -1 & : \ w^T x + b \ < 0 \end{cases}$$

Where $\hat{y}$ is the predicted label of a data point.

Kernel Trick

When data is not linearly separable, SVM uses kernel functions:

- Linear Kernel
- Polynomial Kernel
- RBF (Radial Basis Function) Kernel

4.<u>Algorithm Limitations</u>

1.Computationally expensive for large datasets
2.Sensitive to choice of kernel
3.Requires feature scaling
4.Hard to interpret
5.Poor performance if classes overlap heavily

5.<u>Methodology / Workflow</u>

1️⃣Load Kaggle dataset
2️⃣Data cleaning
3️⃣Feature scaling (important for SVM)
4️⃣Train-test split (80-20)
5️⃣Train SVM model
6️⃣Evaluate performance
7️⃣Hyperparameter tuning

WorkFlow :

Load Dataset
↓
Preprocessing
↓
Feature Scaling
↓
Train-Test Split
↓
Train SVM
↓
Prediction
↓
Evaluation
↓
Hyperparameter Tuning

6.<u>Performance Analysis</u>

Evaluation metrics used:

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1 Score

Interpretation:

- High accuracy indicates good separation of classes
- Balanced precision and recall indicates robust model

7.<u>Hyperparameter Tuning</u>

Important SVM hyperparameters:

Parameter	Meaning
C	Regularization parameter
kernel	Type of kernel
gamma	Influence of single training example

Role of C

Small C → Larger margin, more regularization
Large C → Smaller margin, less regularization

Role of gamma (RBF kernel)

Small gamma → Smooth decision boundary
Large gamma → Complex boundary

GridSearchCV is used to find the best combination.